

## BGFNET: SUPPLEMENTARY MATERIALS

### 1. LOSS FUNCTION

BGF-Net is trained with a multi-output segmentation loss and three background-aware auxiliary objectives. The segmentation loss supervises the final prediction  $P_1$  and the auxiliary predictions  $P_2$ ,  $P_3$ , and  $P_4$ . The auxiliary objectives supervise foreground-background allocation, background semantic prediction, and cross-domain background feature alignment, respectively.

#### 1.1. Multi-Output Segmentation Loss

Let  $Y$  denote the ground-truth foreground mask. Following the weighted structure loss, a pixel-wise weight map is first computed as

$$W = 1 + 5 |\text{AvgPool}_{31}(Y) - Y|, \quad (1)$$

where  $\text{AvgPool}_{31}(\cdot)$  denotes average pooling with a  $31 \times 31$  kernel. The weight map assigns larger weights to pixels around lesion boundaries.

For a prediction logit  $P$ , its probability map is denoted by  $\hat{Y} = \sigma(P)$ . The weighted binary cross-entropy loss and weighted IoU loss are defined as

$$\mathcal{L}_{\text{wbce}} = \frac{\sum_p W^{(p)} \text{BCE}(P^{(p)}, Y^{(p)})}{\sum_p W^{(p)}}, \quad (2)$$

$$\mathcal{L}_{\text{wiou}} = 1 - \frac{\sum_p W^{(p)} \hat{Y}^{(p)} Y^{(p)} + 1}{\sum_p W^{(p)} (\hat{Y}^{(p)} + Y^{(p)} - \hat{Y}^{(p)} Y^{(p)}) + 1}. \quad (3)$$

The structure loss is given by

$$\mathcal{L}_{\text{str}}(P, Y) = \mathcal{L}_{\text{wbce}} + \mathcal{L}_{\text{wiou}}. \quad (4)$$

The multi-output segmentation loss is formulated as

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{str}}(P_1, Y) + \lambda_{\text{aux}} \sum_{i=2}^4 \mathcal{L}_{\text{str}}(P_i, Y), \quad (5)$$

where  $\lambda_{\text{aux}}$  controls the contribution of the auxiliary decoder predictions and is set to 0.2 by default. All predictions are resized to the input-image resolution before computing  $\mathcal{L}_{\text{seg}}$ .

### 1.2. Reliable Foreground-Background Supervision

Direct supervision near polyp boundaries can be unreliable because of annotation ambiguity and severe foreground-background similarity. Therefore, we construct reliable foreground and background regions by eroding and dilating the ground-truth mask:

$$\begin{aligned} R^{\text{fg}} &= \text{Erode}(Y, r), \\ R^{\text{bg}} &= 1 - \text{Dilate}(Y, r), \end{aligned} \quad (6)$$

where  $\text{Erode}(\cdot, r)$  and  $\text{Dilate}(\cdot, r)$  denote morphological erosion and dilation with radius  $r = 5$ , respectively. The valid supervision region is defined as

$$R^{\text{valid}} = \text{clip}(R^{\text{fg}} + R^{\text{bg}}, 0, 1), \quad (7)$$

where  $\text{clip}(\cdot, 0, 1)$  truncates its input to the interval  $[0, 1]$ . Consequently, pixels within the uncertain boundary band are excluded from the background-aware supervision.

At each BGA stage, the allocation masks  $M_i^{\text{fg}}$  and  $M_i^{\text{bg}}$  are supervised within the reliable region. All masks are resized to the corresponding feature resolution by nearest-neighbor interpolation. The allocation loss is defined as

$$\mathcal{L}_{\text{mask}} = -\frac{1}{3} \sum_{i=2}^4 \frac{\sum_p R_i^{\text{valid},(p)} \left[ Y_i^{(p)} \log M_i^{\text{fg},(p)} + (1 - Y_i^{(p)}) \log M_i^{\text{bg},(p)} \right]}{\sum_p R_i^{\text{valid},(p)} + \varepsilon}. \quad (8)$$

where  $i = 2, 3, 4$  indexes the BGA stages,  $p$  indexes spatial locations,  $\varepsilon = 10^{-6}$  is a numerical constant, and  $Y_i$  and  $R_i^{\text{valid}}$  denote the ground-truth mask and valid-region mask resized to the resolution of stage  $i$ , respectively.

### 1.3. Background Semantic Supervision

At each BGA stage, the background feature  $B_i$  is mapped to a background logit  $G_i^{\text{bg}}$ , and the corresponding background probability map is obtained by

$$Q_i^{\text{bg}} = \sigma(G_i^{\text{bg}}). \quad (9)$$

The background branch is supervised using the background target  $1 - Y_i$  within the reliable region:

$$\mathcal{L}_{\text{bg-sem}} = \frac{1}{3} \sum_{i=2}^4 \left[ \mathcal{L}_{\text{bce}} \left( G_i^{\text{bg}}, 1 - Y_i, R_i^{\text{valid}} \right) + \lambda_{\text{dice}} \mathcal{L}_{\text{dice}} \left( Q_i^{\text{bg}}, 1 - Y_i, R_i^{\text{valid}} \right) \right]. \quad (10)$$

where  $i = 2, 3, 4$  indexes the BGA stages,  $G_i^{\text{bg}}$  denotes the background logit, and  $Q_i^{\text{bg}}$  denotes the corresponding background probability map. The functions  $\mathcal{L}_{\text{bce}}(\cdot)$  and  $\mathcal{L}_{\text{dice}}(\cdot)$  denote binary cross-entropy with logits and Dice loss computed only over valid pixels, respectively. We set  $\lambda_{\text{dice}} = 1.0$ . This objective encourages the background branch to capture stable non-polyp structures while avoiding ambiguous boundary regions.

#### 1.4. Background Invariance Loss

To improve robustness under domain shifts, BGF-Net aligns reliable background representations from the third and fourth BGA stages. For each sample from source domain  $d$ , the background feature is pooled over the reliable background region and then  $\ell_2$  normalized:

$$\mathbf{z}_{i,n}^d = \text{Norm} \left( \text{Pool} \left( B_{i,n}^d, R_{i,n}^{\text{bg},d} \right) \right), \quad (11)$$

where  $i$  indexes the BGA stage,  $n$  indexes a sample from source domain  $d$ ,  $R_{i,n}^{\text{bg},d}$  denotes the reliable background mask resized to the resolution of  $B_{i,n}^d$ ,  $\text{Pool}(\cdot)$  denotes masked global average pooling, and  $\text{Norm}(\cdot)$  denotes  $\ell_2$  normalization. Let  $\mathcal{Z}_i^d$  denote the set of normalized background vectors from source domain  $d$  at stage  $i$ .

For the Kvasir-SEG and CVC-ClinicDB source domains, the background invariance loss is calculated using multi-kernel MMD:

$$\mathcal{L}_{\text{inv}} = \frac{1}{|\mathcal{S}_{\text{valid}}|} \sum_{i \in \mathcal{S}_{\text{valid}}} \text{MMD}_{\text{MK}}^2 \left( \mathcal{Z}_i^a, \mathcal{Z}_i^b \right), \quad (12)$$

where  $a$  and  $b$  denote the two source domains, and  $\mathcal{S}_{\text{valid}}$  contains valid stages selected from the third and fourth BGA stages. A stage is included only when both domains contain at least two samples. The multi-kernel MMD is computed by averaging Gaussian kernels with bandwidths  $\Sigma = \{0.5, 1.0, 2.0, 4.0\}$ . This loss encourages background representations from different domains to share similar feature distributions while preserving foreground discrimination through the segmentation objective.

#### 1.5. Overall Objective

The main training objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_{\text{bg}} (\mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{bg-sem}}) + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}}, \quad (13)$$

where  $\lambda_{\text{bg}}$  jointly controls the contributions of the foreground-background allocation loss and background semantic loss, while  $\lambda_{\text{inv}}$  controls the contribution of the background invariance loss. Unless otherwise stated, we set  $\lambda_{\text{bg}} = 0.60$  and  $\lambda_{\text{inv}} = 0.20$ .

**Table 1.** Detailed comparison with SOTA methods on the Kvasir-SEG and CVC-ClinicDB datasets. The best and second-best available results are highlighted in **bold** and underlined, respectively. EFA-Net did not report max  $E_m$  or  $MAE$ .

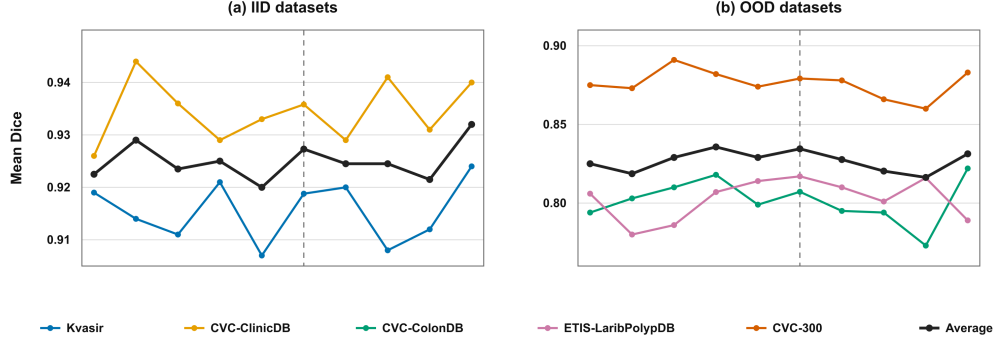
| Method           | Kvasir-SEG   |              |              |              |              |              | CVC-ClinicDB |              |              |              |              |              |
|------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                  | $mDic$       | $mIoU$       | $F_\beta^w$  | $S_\alpha$   | max $E_m$    | $MAE$        | $mDic$       | $mIoU$       | $F_\beta^w$  | $S_\alpha$   | max $E_m$    | $MAE$        |
| UNet (2015)      | 0.818        | 0.746        | 0.794        | 0.858        | 0.893        | 0.055        | 0.823        | 0.755        | 0.811        | 0.889        | 0.954        | 0.019        |
| PraNet (2020)    | 0.896        | 0.838        | 0.880        | 0.913        | 0.944        | 0.030        | 0.900        | 0.849        | 0.896        | 0.937        | 0.979        | 0.009        |
| Polyp-PVT (2021) | <u>0.918</u> | <u>0.868</u> | <b>0.915</b> | 0.924        | <u>0.970</u> | <b>0.021</b> | <u>0.935</u> | <u>0.885</u> | <u>0.935</u> | 0.942        | 0.989        | <u>0.007</u> |
| EFA-Net (2025)   | 0.914        | 0.861        | 0.906        | <u>0.929</u> | –            | –            | 0.919        | 0.871        | 0.916        | 0.943        | –            | –            |
| FMCA-Net (2025)  | 0.906        | 0.851        | 0.902        | 0.917        | 0.962        | 0.026        | 0.930        | 0.881        | 0.928        | <u>0.946</u> | <u>0.992</u> | <u>0.007</u> |
| PraNet-V2 (2026) | 0.915        | 0.861        | 0.904        | 0.925        | 0.960        | 0.023        | 0.931        | 0.881        | 0.928        | 0.944        | 0.986        | 0.008        |
| BGF-Net (Ours)   | <b>0.921</b> | <b>0.870</b> | <u>0.914</u> | <b>0.936</b> | <b>0.982</b> | <u>0.022</u> | <b>0.938</b> | <b>0.891</b> | <b>0.937</b> | <b>0.957</b> | <b>0.995</b> | <b>0.006</b> |

**Table 2.** Detailed comparison with SOTA methods on the CVC-ColonDB, CVC-300, and ETIS datasets. The best and second-best available results are highlighted in **bold** and underlined, respectively. EFA-Net did not report max  $E_m$  or  $MAE$ .

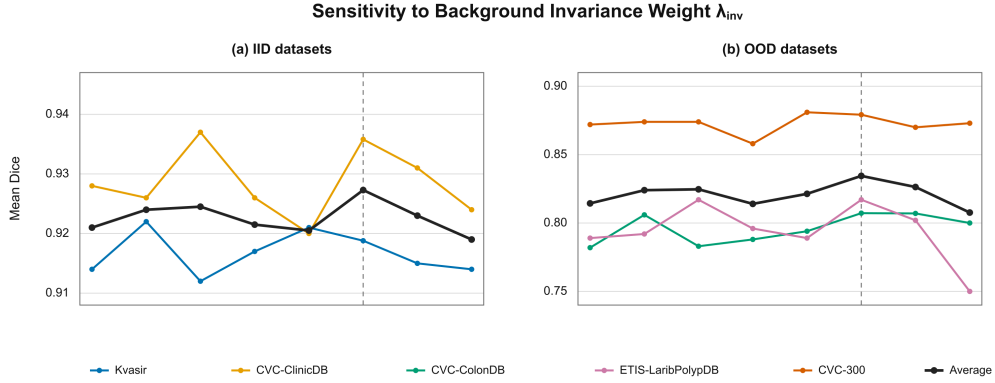
| Method           | CVC-ColonDB  |              |              |              |              |              | CVC-300      |              |              |              |              |              | ETIS         |              |              |              |              |              |
|------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                  | $mDic$       | $mIoU$       | $F_\beta^w$  | $S_\alpha$   | max $E_m$    | $MAE$        | $mDic$       | $mIoU$       | $F_\beta^w$  | $S_\alpha$   | max $E_m$    | $MAE$        | $mDic$       | $mIoU$       | $F_\beta^w$  | $S_\alpha$   | max $E_m$    | $MAE$        |
| UNet (2015)      | 0.512        | 0.444        | 0.498        | 0.712        | 0.776        | 0.061        | 0.710        | 0.627        | 0.684        | 0.843        | 0.875        | 0.022        | 0.398        | 0.335        | 0.366        | 0.684        | 0.740        | 0.036        |
| PraNet (2020)    | 0.713        | 0.642        | 0.699        | 0.820        | 0.872        | 0.043        | 0.874        | 0.799        | 0.843        | 0.925        | 0.972        | 0.010        | 0.630        | 0.568        | 0.601        | 0.793        | 0.841        | 0.031        |
| Polyp-PVT (2021) | 0.793        | 0.711        | 0.780        | 0.858        | 0.908        | <b>0.032</b> | 0.893        | 0.825        | <u>0.878</u> | 0.930        | 0.976        | 0.009        | 0.786        | 0.703        | <u>0.753</u> | 0.871        | 0.915        | 0.018        |
| EFA-Net (2025)   | 0.774        | 0.696        | 0.753        | 0.855        | –            | –            | 0.894        | <u>0.830</u> | <u>0.878</u> | <b>0.941</b> | –            | –            | 0.749        | 0.670        | 0.698        | 0.858        | –            | –            |
| FMCA-Net (2025)  | <u>0.795</u> | <u>0.714</u> | <u>0.782</u> | <u>0.861</u> | 0.921        | <b>0.032</b> | 0.892        | 0.827        | 0.872        | 0.933        | 0.978        | <u>0.008</u> | <u>0.791</u> | <u>0.712</u> | <u>0.753</u> | <u>0.877</u> | <u>0.936</u> | 0.017        |
| PraNet-V2 (2026) | 0.767        | 0.687        | 0.752        | 0.845        | 0.892        | 0.036        | <b>0.899</b> | <b>0.831</b> | <b>0.885</b> | <u>0.940</u> | 0.976        | <b>0.007</b> | 0.764        | 0.687        | 0.730        | 0.865        | 0.888        | <u>0.015</u> |
| BGF-Net (Ours)   | <b>0.802</b> | <b>0.726</b> | <b>0.787</b> | <b>0.869</b> | <b>0.951</b> | <b>0.032</b> | <u>0.896</u> | 0.827        | 0.874        | 0.937        | <b>0.988</b> | <b>0.007</b> | <b>0.834</b> | <b>0.761</b> | <b>0.806</b> | <b>0.908</b> | <b>0.968</b> | <b>0.012</b> |

**Table 3.** Sensitivity to the reliable-region radius  $r$  under  $\lambda_{bg} = 0.60$  and  $\lambda_{inv} = 0.20$ . Avg. denotes the average over five datasets.

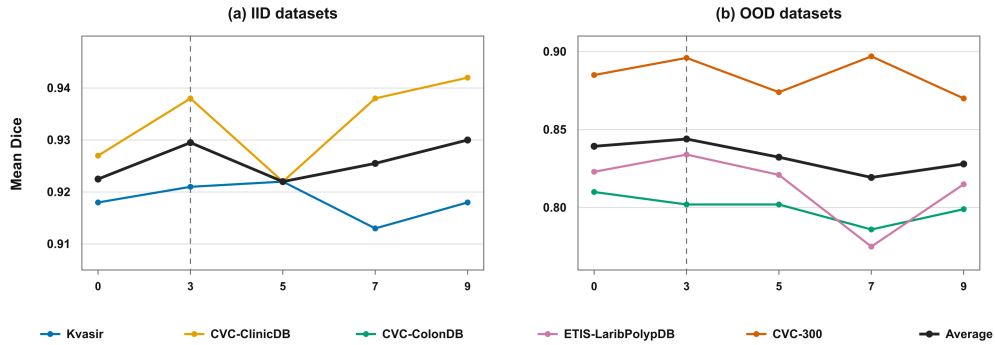
| $r$ | Kvasir-SEG |        | CVC-ClinicDB |        | CVC-ColonDB |        | CVC-300 |        | ETIS   |        | Avg.         |              |
|-----|------------|--------|--------------|--------|-------------|--------|---------|--------|--------|--------|--------------|--------------|
|     | $mDic$     | $mIoU$ | $mDic$       | $mIoU$ | $mDic$      | $mIoU$ | $mDic$  | $mIoU$ | $mDic$ | $mIoU$ | $mDic$       | $mIoU$       |
| 0   | 0.918      | 0.871  | 0.927        | 0.881  | 0.810       | 0.730  | 0.885   | 0.817  | 0.823  | 0.753  | 0.873        | 0.810        |
| 3   | 0.921      | 0.870  | 0.938        | 0.891  | 0.802       | 0.726  | 0.896   | 0.827  | 0.834  | 0.761  | <b>0.878</b> | <b>0.815</b> |
| 5   | 0.922      | 0.873  | 0.922        | 0.875  | 0.802       | 0.726  | 0.874   | 0.806  | 0.821  | 0.751  | 0.868        | 0.806        |
| 7   | 0.913      | 0.862  | 0.938        | 0.893  | 0.786       | 0.713  | 0.897   | 0.831  | 0.775  | 0.708  | 0.862        | 0.801        |
| 9   | 0.918      | 0.866  | 0.942        | 0.898  | 0.799       | 0.720  | 0.870   | 0.802  | 0.815  | 0.734  | 0.869        | 0.804        |



**Fig. 1.** Sensitivity analysis of the background supervision weight  $\lambda_{bg}$ , where  $\lambda_{mask} = \lambda_{sem} = \lambda_{bg}$ . From left to right, the horizontal-axis points correspond to  $\lambda_{bg} \in \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}$ . The dashed line indicates the selected setting  $\lambda_{bg} = 0.60$ .



**Fig. 2.** Sensitivity analysis of the background-invariance weight  $\lambda_{inv}$  for MMD regularization. From left to right, the horizontal-axis points correspond to  $\lambda_{inv} \in \{0, 0.03, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ . The dashed line indicates the selected setting  $\lambda_{inv} = 0.20$ .



**Fig. 3.** Sensitivity analysis of the reliable-region radius  $r$  under  $\lambda_{bg} = 0.60$  and  $\lambda_{inv} = 0.20$ . From left to right, the horizontal-axis points correspond to  $r \in \{0, 3, 5, 7, 9\}$ . The dashed line indicates the selected setting  $r = 3$ .